Software Requirements Specification

For

AI STUDY COMPANION FOR UNITEN STUDENTS

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Contents

1. Introduction & Purpose 4

Use case Diagram 4

2. Project Goal & Scope 5

2.1 Project Goal 5

2.2 Scope of the System 5

Flowchart: 7

3. Stakeholders & Users 7

4. Functional Requirements 8

4.1 User Authentication 8

4.2 Document Upload & Processing 8

4.3 Note Summarization 8

4.4 Quiz Generation 8

4.5 Quiz Taking 8

4.6 Writing Improvement 8

4.7 Task Scheduling 8

4.8 UNITEN Academic Q&A 9

4.9 Admin Functions 9

Table for Functional Requirement 9

5. Non-Functional Requirements 10

5.1 Performance 10

5.2 Security 10

5.3 Usability 10

5.4 Reliability 10

5.5 Compatibility 10

5.6 Maintainability 10

Table for Non-Functional Requirements 10

6. Interfaces 11

6.1 User Interface (UI) 11

6.2 Admin Interface 12

6.3 System Interface (Backend) 13

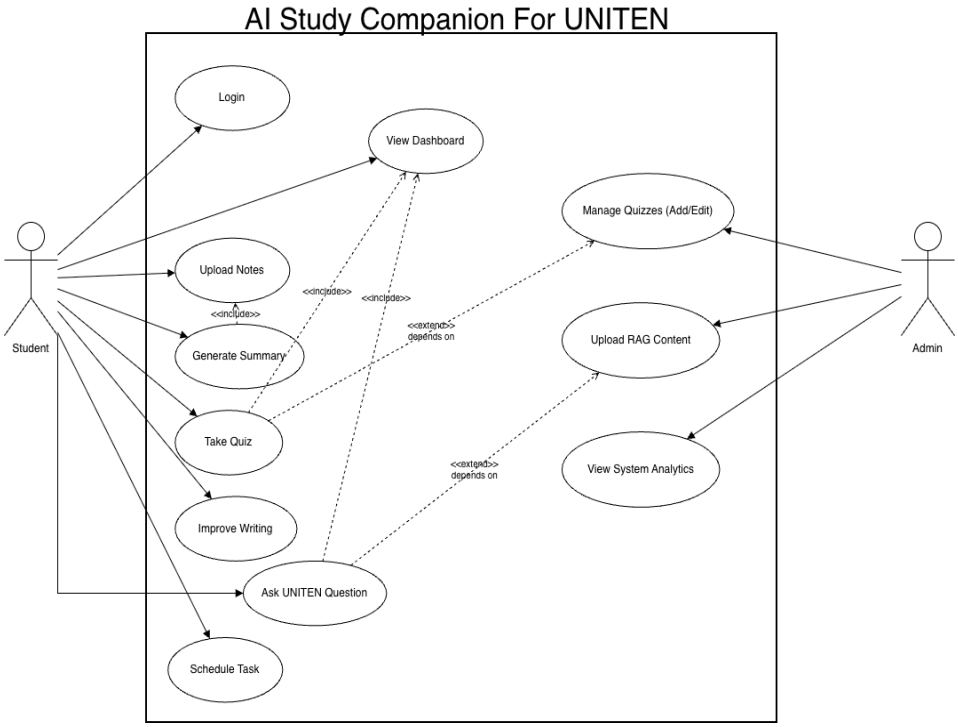
1. Introduction & Purpose

The purpose of this Software Requirements Specification (SRS) document is to define and describe the functional and non-functional requirements of the AI Study Companion for UNITEN Students, a web-based intelligent academic assistant. The document outlines the system goals, scope, targeted stakeholders, functionalities, performance expectations, and interface descriptions.

This SRS serves as the foundation for system design, development, and validation. It ensures that developers, supervisors, and stakeholders share a common understanding of the system's expected behavior, constraints, and goals.

The AI Study Companion is intended to help UNITEN students improve academic productivity by integrating note summarization, quiz generation, writing enhancement, task scheduling, and UNITEN-specific academic Q&A into a single unified platform powered by Natural Language Processing (NLP) and a fine-tuned Large Language Model (LLM).

Use case Diagram



2. Project Goal & Scope

2.1 Project Goal

The main goal of this project is to develop an AI-powered academic assistant that helps UNITEN students manage coursework more efficiently by providing personalized support through automation and intelligent content generation. The system aims to serve as an all-in-one study portal that enhances learning organization and academic performance.

2.2 Scope of the System

Functional Scope

The system will include the following core functionalities:

- Note Summarization
  - Upload notes (PDF, DOCX, TXT)
  - Generate concise, readable summaries
  - Extract key points and concepts
- Quiz Generation & Quiz Taking
  - Automatically generate MCQs/short-answer quizzes from notes
  - Students can take quizzes and receive instant grading
  - Admin can create/edit/manage quizzes
- Writing Enhancement
  - Improve grammar, clarity, vocabulary, and academic tone
  - Compare original and improved versions
- Task Scheduling & Reminders
  - Add/edit/delete academic tasks
  - Set deadlines and prioritized reminders
  - Sync with academic schedule
- UNITEN Academic Q&A (RAG-Powered)
  - Answer UNITEN-specific questions (policies, procedures, rules)
  - Retrieve information from UNITEN documents uploaded by admins

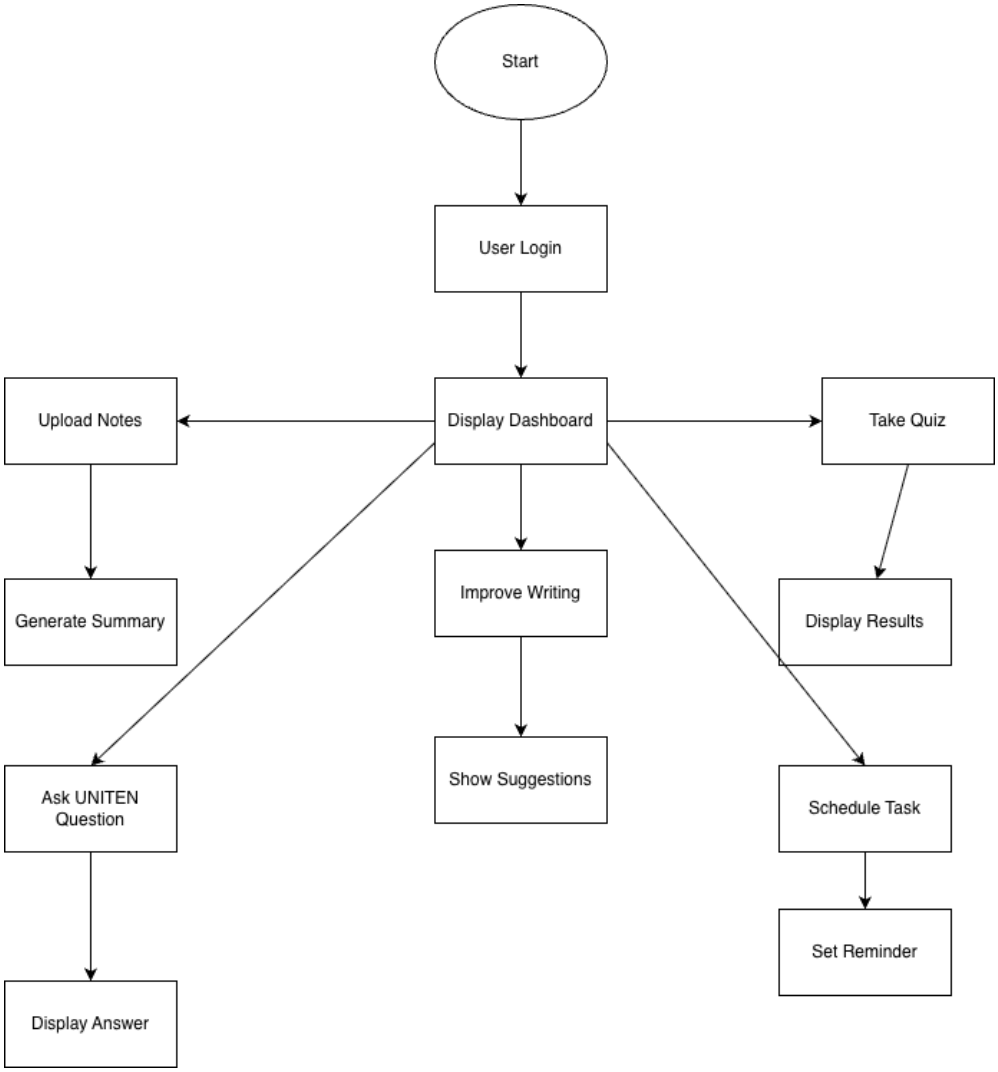
User Scope

- Primary Users:
    - UNITEN students from all faculties
    - They interact with summarization, quiz, writing, scheduling, and AI-Q&A features.
- Secondary Users:
    - UNITEN academic staff (lecturers/admins)
    - They manage system content, quizzes, and RAG documents.

Technical Scope

- Web-based platform accessible on mobile/desktop
- Uses LLM + RAG pipeline for UNITEN-specific responses
- Uses Supabase for database, storage, vector embeddings
- JSON-based APIs via Next.js backend
- Supports secure authentication and data protection

Flowchart:



3. Stakeholders & Users

Stakeholder	Role	Interest
Students	Primary Users	Need accurate summaries, quizzes, writing help, task scheduling
Lecturers/Admins	Secondary Users	Upload RAG documents, manage quizzes and student data
FYP Supervisor	Oversees project	Ensures academic alignment & quality

Developers	System creators	Implement all components and maintain system

4. Functional Requirements

4.1 User Authentication

- The system shall allow users to register and log in using UNITEN email.
- The system shall provide secure authentication using encrypted passwords.

4.2 Document Upload & Processing

- The system shall allow uploading of PDF, DOCX, and TXT files.
- The system shall convert uploaded files into text for processing.

4.3 Note Summarization

- The system shall generate concise summaries from uploaded notes.
- The system shall display summaries and store them for later access.

4.4 Quiz Generation

- The system shall automatically generate quizzes from uploaded notes.
- The system shall support MCQ and short-answer question formats.

4.5 Quiz Taking

- The system shall allow students to attempt quizzes and receive scores.
- The system shall save student quiz results for history and review.

4.6 Writing Improvement

- The system shall allow students to paste or upload text for improvement.
- The system shall provide grammar, clarity, tone, and vocabulary suggestions.
- The system shall show the improved version alongside the original.

4.7 Task Scheduling

- The system shall allow creating/editing/deleting study tasks.
- The system shall provide reminder notifications for upcoming deadlines.

4.8 UNITEN Academic Q&A

- The system shall answer UNITEN-specific questions using the RAG pipeline.
- The system shall show citations to source documents.

4.9 Admin Functions

- Admin shall upload UNITEN policy documents for RAG indexing.
- Admin shall create, edit, and publish quizzes.
- Admin shall view system analytics (usage, uploads, quiz stats).

Table for Functional Requirement

ID	Requirement Description	Priority
FR1	The system shall allow users to register and log in using their UNITEN email and a password.	High
FR2	The system shall allow users to upload documents (PDF, DOCX, TXT) for processing.	High
FR3	The system shall automatically generate a concise summary from an uploaded document or pasted text.	High
FR4	The system shall automatically create multiple-choice and short-answer quizzes based on the content of uploaded notes.	High
FR5	The system shall analyze user-provided text and suggest improvements for grammar, spelling, clarity, and academic tone.	High
FR6	The system shall provide accurate answers to questions about UNITEN's academic policies, courses, and procedures.	High
FR7	The system shall allow users to create, view, edit, and delete academic tasks and set deadlines.	Medium
FR8	The system shall send browser or email notifications for The upcoming task deadlines.	Medium
FR9	system shall accept voice commands for core functions like asking questions or starting a summary.	Low

5. Non-Functional Requirements

5.1 Performance

- The system shall load all major pages in under 3 seconds.
- Summaries for a 10-page document must be generated within 15 seconds.

5.2 Security

- User passwords must be stored using strong encryption (e.g., hashing).
- Only authorized users (admins) may upload RAG documents.
- Uploaded academic documents must remain private and protected.

5.3 Usability

- The user interface shall be simple, intuitive, and beginner-friendly.
- Features must be easily discoverable without a tutorial.

5.4 Reliability

- System availability must be at least 99% during peak academic hours.
- The system must gracefully handle failed uploads or model errors.

5.5 Compatibility

- The system must be responsive on mobile, tablet, and desktop.
- Should function properly across major browsers (Chrome, Edge, Safari).

5.6 Maintainability

- The system shall support easy updates and addition of new modules.
- Codebase should follow modular and reusable structure.

Table for Non-Functional Requirements

ID	Requirement Description	Category
NFR1	The system shall load web pages in less than 3 seconds and generate a summary for a 10-page document within 15 seconds.	Performance

NFR2	The user interface shall be simple and easy to learn. A new user should be able to complete core tasks without a tutorial.	Usability
NFR3	The system shall be accessible via web browsers on desktop computers, tablets, and mobile phones without loss of functionality.	Compatibility
NFR4	User passwords shall be stored in an encrypted format. User data and uploaded documents shall be kept private and secure.	Security
NFR5	The system shall be available for use 99% of the time during peak hours (8 AM to 10 PM).	Reliability
NFR6	The system shall be built in a way that makes it easy to add new features, like a grade calculator, in the future.	Maintainability

6. Interfaces

6.1 User Interface (UI)

Login Screen

- Fields: email, password
- Role-based redirection:
- Students → Student Dashboard
- Admin/Lecturer → Admin Dashboard

Student Dashboard

- Displays:
    - Recent notes
    - Upcoming tasks
    - Quick access to quizzes
    - AI Q&A panel
    - Note Summarizer Interface

    - Shows uploaded documents list
    - Provides: upload button, search bar, view summary page
- Quiz Interface
    - Quiz list view (searchable)
    - Quiz attempt screen (timer, questions, navigation)
    - Results screen (score + feedback)
- Writing Improvement Page
    - Editor for pasting or typing text
    - Side-by-side “Original vs Improved” results
- AI Q&A Chat Interface
    - Chat window
    - Response with citations
    - History of questions
- Task Scheduler
    - Add/edit/delete task forms
    - Calendar and reminder settings

6.2 Admin Interface

Admin Dashboard

- Overview of:
    - Total students
    - Uploaded RAG documents
    - Active quizzes
    - Pending queries
    - RAG Document Upload Page
    - Upload form
    - Metadata form (department, course, content type)
    - Status/log view
- Quiz Management
    - Create quiz (manual or AI-generated)
    - Edit quiz

    - Publish/unpublish quiz
- User Management
    - Approve student accounts
    - Edit or deactivate accounts

6.3 System Interface (Backend)

- RESTful/JSON API endpoints via Next.js
- AI endpoints (summarization, Q&A, writing improvements)
- Database connection via Supabase
- Vector search integration via pgvector